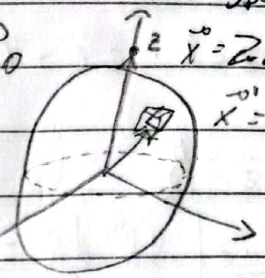
Lista 3 - Eletromagnetismo

Elias Vitor da Silva

7 - $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{x}')(\vec{x}-\vec{x}')}{|\vec{x}-\vec{x}'|^3}dV'$ $\qquad \rho(\vec{x}') = \rho_0$



$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho_0(z\hat{a}_z - R'_{a_R})}{|z\hat{a}_z - R'\hat{a}_R|^3}dV \Rightarrow \frac{\rho_0}{4\pi\varepsilon_0}\int_0^R\int_0^\pi\int_0^{2\pi} \frac{z\hat{a}_z - R\hat{a}_R}{|z\hat{a}_z - R\hat{a}_R|}dV' \qquad dV = R^2 \operatorname{sen}\theta' dR' d\theta' d\phi$$

$$= \int_0^a\int_0^\pi\int_0^{2\pi} \frac{z\hat{a}_R - R(\operatorname{sen}\theta\hat{a}_R + \cos\theta\hat{a}_z)}{(z^2+R^2-2Rz\cos\theta)^{3/2}} R^2 \operatorname{sen}\theta\, dR' d\theta' d\phi' \Rightarrow \int_0^a R^2 dR \int_0^\pi \frac{\operatorname{sen}\theta}{z^2+R^2-2Rz\cos\theta} d\theta' \int_0^{2\pi} z\hat{a}_z - R\operatorname{sen}\theta\cos\theta$$

$$= \int_0^a R^2 dR \int_0^\pi \frac{\operatorname{sen}\theta'}{(z^2+R^2-2Rz\cos\theta')^{3/2}} d\theta' \int_0^{2\pi} z_{a_z} - R\cos\theta\hat{a}_z - R\operatorname{sen}\theta\hat{a}_R\, d\phi$$

$a_R = \operatorname{sen}\theta\hat{a}_R + \cos\theta a_z$

$a_R = \operatorname{sen}\phi\hat{a}_y + \cos\phi\hat{a}_x$

$$= \int_0^a R^2 dR' \int_0^\pi \frac{\operatorname{sen}\theta'}{(z^2+R^2-2Rz\cos\theta)^{3/2}} d\theta' \int_0^{2\pi} z\hat{a}_z - R\cos\hat{a}_z - R\operatorname{sen}\theta \int_0^{2\pi} \hat{a}_R d\phi \Rightarrow \int_0^{2\pi} \operatorname{sen}\phi\hat{a}_y + \cos\phi\hat{a}_x d\phi$$

$$-\cos\phi + \operatorname{sen}\phi\Big|_0^{2\pi} \quad \cancel{\operatorname{sen}(2\pi)}^{0} - \cos(2\pi) - (\cancel{\operatorname{sen}(0)}^{0} - \cos(0)) = -1 + 1 = \boxed{0}$$

$$= \int_0^a R^2 dR' \int_0^\pi \frac{(z\hat{a}_z - R'\cos\theta'\hat{a}_z)\operatorname{sen}\theta' d\theta}{(z^2+R^2-2Rz\cos\theta)^{3/2}} \cdot \cancel{\int_0^{2\pi} d\phi}^{2\pi} \Rightarrow \frac{\rho_0}{2\varepsilon_0}\iint \ldots d\theta'$$

$$\int_0^a R^2 dR' \int_0^\pi \frac{2z^2 - z^2 + R^2 - u}{2Rz \cdot 2z \cdot u^{3/2}} du \Rightarrow \frac{1}{4Rz^2}\int_{u_1}^{u_2} (z^2+R^2-u)\, u^{-3/2} du$$

$u = z^2 + R^2 - 2Rz\cos\theta$

$du = 2Rz\operatorname{sen}\theta d\theta$

$\operatorname{sen}\theta d\theta = \frac{du}{2Rz}$

$\cos\theta = \frac{z^2+R^2-u}{2Rz}$

$u_1 = (z-R)^2$

$u_2 = (z+R)^2$

$$\Rightarrow z^2+R^2 \int_{u_1}^{u_2} u^{-3/2} du - \int_{u_1}^{u_2} u^{-1/2} du$$

$$\Rightarrow \sqrt{(R+z)^2} - \sqrt{(R-z)^2} - (z^2+R^2)\cdot\left(\frac{1}{\sqrt{(z+R)^2}} - \frac{1}{\sqrt{(z-R)^2}}\right)$$

Lembrar: $R + z - |R - z| - (z^2 - R^2)\cdot\left(\frac{1}{z+R} - \frac{1}{|z-R|}\right)$

Para $z > R = 2R \cdot (z^2 - R^2) \cdot \frac{-2R}{z^2 - R^2} = 4R$ $\quad \dot{g}(R) = \begin{cases} 4R, & z > R \\ 0, & z < R \end{cases}$

Para $z < R$: $2z - (z^2 - R^2) \cdot \frac{2z}{z^2 - R^2} = 0$

Para $z > R$

$$\frac{P_0 \hat{a}_z}{4\varepsilon_0 z^2} \int_0^a 4R'^2 \, dR' = \frac{P_0 \hat{a}_z}{4\varepsilon_0 z^2} \cdot \frac{4R'^3}{3} \Big|_0^a = \boxed{\frac{P_0 a^3}{3\varepsilon_0 z^2} \hat{a}_z}$$

Para $z < R$:

$$\frac{P_0 \hat{a}_z}{4\varepsilon_0 z^2} \int_0^a R' \, dR' = \int_0^z 4R'^2 \, dR' + \int_z^a 0 \, dR' \Rightarrow \boxed{\frac{P_0 z}{3\varepsilon_0} \hat{a}_z}$$

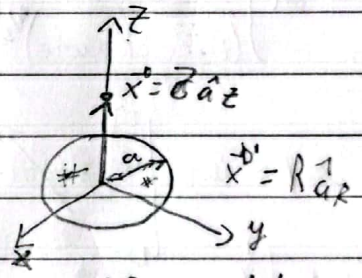
2- $$\Phi = \frac{1}{4\pi\varepsilon_0} \int_\Sigma \frac{P(\vec{x}')}{|\vec{x} - \vec{x}'|} d\Sigma \Rightarrow \frac{P_2}{4\pi\varepsilon_0} \int_0^a \int_0^{2\pi} \frac{r \, dr \, d\phi}{|z\hat{a}_z - r\hat{a}_r|}$$

$$= \cancel{\int_0^{2\pi} d\phi} \int \frac{r \, dr}{(z^2 + r^2)^{1/2}} \Rightarrow \frac{P_2}{2\varepsilon_0} \int_0^a \frac{r}{(z^2 + r^2)^{1/2}} dr$$

$$\int_0^a \frac{du}{2u^{1/2}} \Rightarrow \frac{1}{\cancel{2}} \cdot \left[\cancel{2} u^{+1/2} \right]_{u_1}^{u_2} = \sqrt{z^2 + r^2} \Big|_0^a$$

$$\sqrt{z^2 + a^2} - \sqrt{z^2} \quad \Big| \Rightarrow \boxed{\frac{P_2 a}{2\varepsilon_0} = \Phi}$$

$|a| \cdot |z| - |z|$



$d\vec{\xi} = r \, dr \, d\phi$

$P(\vec{x}') = P_2$

$u = z^2 + r^2$

$du = 2r \, dr$

$r \, dr = \frac{du}{2}$

$a >> z$

$$E = -\vec{\nabla} \cdot \Phi$$

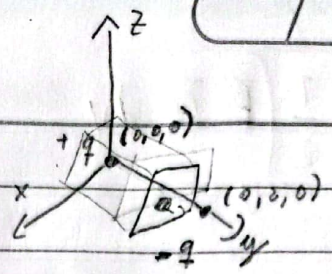
$$E = -\left(\cancel{\frac{d}{dr}}^{0} + \cancel{\frac{1}{r} \frac{d\phi}{d\phi}}^{0} + \frac{d}{dz} \right) \cdot \frac{P_2 \cdot (|a| - |z|)}{2\varepsilon_0} = -\frac{d}{dz} \left(\frac{P_2 \cdot (|a| - |z|)}{2\varepsilon_0} \right)$$

$$\boxed{\frac{P_2}{2\varepsilon_0} \frac{z}{|z|}}$$

tilibra

CamScanner

3- $\oint_{\mathcal{E}} \vec{D} \cdot d\vec{\mathcal{E}} = q$

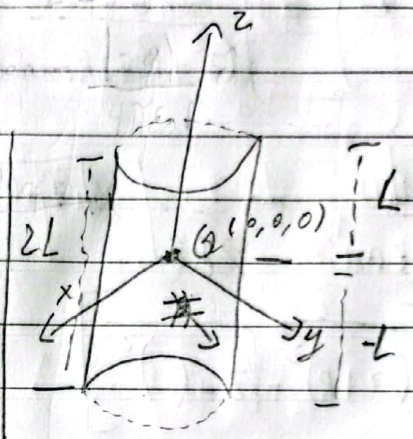


como um cubo possui 6 faces e só queremos o valor do fluxo em apenas 1 das faces temos que o fluxo sobre a superfície

$\{y = 1, -1 \leq x < 1, -1 \leq z < 1\} = \boxed{\frac{q}{6}}$

$d\vec{\mathcal{E}} = dx\,dz\,\hat{a}_y$

4- $\int_{\mathcal{E}} \vec{D} \cdot d\vec{\mathcal{E}} = \int_0^{2\pi}\int_0^{a} \left(\frac{Q}{4\pi R^2}\hat{a}_R\right) \cdot \left(r\,d\phi\,dz\,\hat{a}_r\right)$



$\frac{Q}{4\pi} \cdot \int_0^{2\pi}\int_{-L}^{L} \frac{r \operatorname{sen}\theta}{R^2}\, d\phi\, dz \Rightarrow \int_0^{2\pi}\int_{-L}^{L} \frac{r^2}{(r^2+z^2)\cdot\sqrt{r^2+z^2}}\, d\phi\, dz$

$d\vec{\mathcal{E}} = r\,d\phi\,dz\,\hat{a}_r$

$\vec{D} = \frac{Q}{4\pi R^2}\hat{a}_R$

$|a_r| \cdot |a_R| \cos\theta = \operatorname{sen}\theta$

$\frac{Q}{4\pi}\int_0^{2\pi} d\phi \int_{-L}^{L} \frac{r^2}{(r^2+z^2)^{3/2}}\, dz$

$\frac{Q}{2}\int_{-L}^{L} \frac{r^2}{(r^2+z^2)^{3/2}}\, dz \Rightarrow r^2 \int_{-L}^{L} \frac{1}{(r^2+z^2)^{3/2}}\, dz$

$u = r\tan(u)$

$du = r\sec(u)$

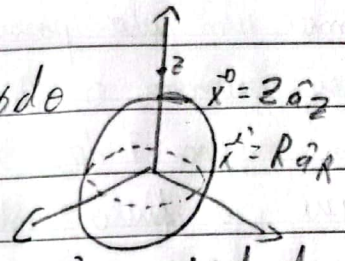
$\int_{\mu_1}^{\mu_2} \frac{1}{r^2 \sec(u)}\, du$

$\frac{r^2}{r^2} \cdot \frac{1}{r^2} \int \frac{1}{\sec(u)}\, du$

$\int_{\mathcal{E}} \vec{D} \cdot d\vec{\mathcal{E}} = \boxed{\frac{Q\,l}{(l^2+r^2)^{1/2}}\,\hat{a}_r}$

5- $W = \frac{1}{2}\int_V \rho\, \Phi\, dv$, $\Phi = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{x}')}{|\vec{x}-\vec{x}'|}\, dv$

$$\Phi = \frac{\rho_0}{4\pi\varepsilon_0}\int_V \frac{R^2 \operatorname{sen}\theta\, dR\, d\phi\, d\theta}{|z\hat{a}_z + R\hat{a}_R|} \Rightarrow \int_0^a\int_0^\pi\int_0^{2\pi} \frac{R^2 \operatorname{sen}\theta\, dR\, d\phi\, d\theta}{|z+R|}$$



$dv = R^2 \operatorname{sen}\theta\, dR\, d\phi\, d\theta$

$\rho(\vec{x}') = \rho_0$

$$\Rightarrow \int_0^a R^2 dR \int_0^{2\pi} d\phi \int_0^\pi \frac{\operatorname{sen}\theta}{(z^2+R^2+2Rz\cos\theta)^{1/2}}\, d\theta$$

$$\int_0^\pi \frac{du}{2Rz \cdot u^{1/2}} \Rightarrow \frac{1}{2Rz}\cdot\left[2\sqrt{z^2+R^2+2Rz\cos\theta}\right]_0^\pi$$

$u = z^2+R^2+2Rz\cos\theta$

$du = -2Rz \operatorname{sen}\theta\, d\theta$

$\operatorname{sen}\theta\, d\theta = \frac{du}{+2Rz}$

$\cos(\pi) = -1$

$\cos(0) = 1$

$$\frac{1}{Rz}\left[\sqrt{(z+R)^2} - \sqrt{(z-R)^2}\right]$$

$$\frac{1}{4\pi\varepsilon_0}\int_0^{2\pi} d\phi \int_0^a R^2 \cdot \frac{1}{Rz}\left[\sqrt{(z+R)^2} - \sqrt{(z-R)^2}\right] dR$$

$$\Phi = \frac{\rho_0}{2\varepsilon_0 z}\int_0^a R\cdot\left(z+R-|z-R|\right) \Rightarrow \begin{cases} 2R^2, & z > R \\ 2zR, & z < R \end{cases}$$

$$\Phi = \frac{\rho_0}{2\varepsilon_0 z}\int_0^z 2R^2 dR + \int_z^a 2Rz\, dR \Rightarrow \frac{\rho_0}{2\varepsilon_0 z}\left[\frac{2}{3}z^3 + z(a^2-z^2)\right]$$

$$\Phi \frac{\rho_0}{2\varepsilon_0}\cdot\frac{2z^2}{3} + (a^2-z^2) \Rightarrow \frac{\rho}{2\varepsilon_0}\cdot\frac{3a^2-z^2}{3} = \frac{\rho_0\, 3a^2-z^2}{6\varepsilon_0}$$

$$W_f = \frac{1}{2}\int_V \rho\left[\frac{\rho\, 3a^2-z^2}{6\varepsilon_0}\right] dv \Rightarrow \frac{\rho^2}{12\varepsilon_0}\underbrace{\int d\Omega}_{4\pi}\int_0^a R^2\left[3a^2-R^2\right] dR$$

$$\frac{\rho^2\pi}{3\varepsilon_0}\int_0^a 3aR^2 - R^2 = \frac{\rho^2\pi}{3\varepsilon_0}\cdot\frac{3a^5}{3} - \frac{a^5}{5} \Rightarrow \frac{\rho^2\pi}{3\varepsilon_0}\cdot\frac{5a^5-a^5}{5} \quad \boxed{\frac{4\pi\rho^2 a^5}{15\varepsilon_0}}$$

6- $W_E = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}\,dV \Rightarrow \frac{1}{2}\int_V \varepsilon\,\vec{E}\cdot\vec{E}\,dV = \frac{\varepsilon_0}{2}\int E^2\,dV$

$$E = -\vec{\nabla}\Phi = -\left(\cancel{\frac{d}{dR}}^{\,0} + \cancel{\frac{d}{d\theta}\frac{1}{R}}^{\,0} + \cancel{\frac{1}{R\sin\theta}\frac{d}{d\varphi}}\right)\cdot\frac{P_0[3a^2-R^2]}{6\varepsilon_0} = -\frac{d}{dR}\left(\frac{P_0[3a^2-R^2]}{6\varepsilon_0}\right)$$

$$\vec{E} = \frac{-P_0}{6\varepsilon_0}\,-2R = \frac{P_0 R}{3\varepsilon_0}$$

$$W_E = \frac{\varepsilon_0}{2}\int_V\left(\frac{P_0 R}{3\varepsilon_0}\right)^2 dV = \frac{P_0^2}{18\varepsilon_0}\int_0^a R^4\,dR\underbrace{\int_0^{4\pi} d\Omega}_{4\pi} \Rightarrow \frac{4\pi P_0^2}{18\varepsilon_0}\left[\frac{R^5}{5}\right]_0^a$$

$$\frac{2\pi P_0^2}{9\varepsilon_0}\,\frac{a^5}{5} = \boxed{\frac{2\pi P_0^2 a^5}{45\varepsilon_0}}$$

$\vec{D} = D(R)\,\hat{a}_R$

$\rho = \frac{P_0\pi}{a}e^{-2\pi^3}$

7- $\oint_S \vec{D}\cdot d\vec{S} = \int_V \rho(\vec{v})\,dV$

$$D_R\,4\pi R^2 = \int_0^{2\pi}\int_0^{\pi}\int_0^{R}\frac{P_0\pi}{a}e^{-2\pi^3}\cdot R^2\sin\theta\,d\phi\,d\theta\,dR \Rightarrow \int_0^{4\pi}d\Omega\int_0^R\frac{P_0\pi}{a}e^{-2\pi^3}R^2\,dR$$

$$D_R\,\cancel{4\pi}R^2 = \cancel{4\pi}\frac{P_0\pi}{a}e^{-2\pi^3}\int_0^R R^2\,dR \Rightarrow \frac{P_0\pi}{a}e^{-2\pi^3}\cdot\left[\frac{R^3}{3}\right]_0^R \Rightarrow \frac{P_0\pi}{a}e^{-2\pi^3}\cdot\frac{R^3}{3}$$

$$D_R = \frac{\frac{P_0\pi}{a}e^{-2\pi^3}R}{3} = \boxed{3P\frac{\pi}{a}e^{-2\pi^3}R\ ,\ R\leq a}$$

$$\oint_{\varepsilon_2} \vec{D}\cdot d\vec{\varepsilon}_2 = \int_V \rho(\vec{r})\, dV$$

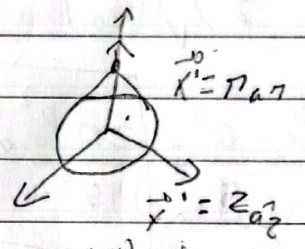
$$D_R\, 4\pi R^2 = \int_0^a \int_0^{4\pi} P_0 \frac{r}{a} e^{-2r^3} R^2\, dR\, d\Omega$$

$$D_R\, 4\pi R^2 = P_0 \frac{r}{a} e^{-2r^3} \cdot \left[4\pi \frac{a^3}{3}\right] \Rightarrow 3P_0 r e^{-2r^3} a^2$$

$$\boxed{D_R = \frac{3P_0 r e^{-2r^3} a^2}{R^2}}$$

8- $W = \frac{1}{2} q\, \Phi(0)$

$$\Phi = \frac{1}{4\pi\varepsilon_0} \int_C \frac{\rho(\vec{x}')\, dl}{|\vec{x} - \vec{x}'|} \Rightarrow \frac{\lambda}{4\pi\varepsilon_0} \int_0^{2\pi} \frac{a\, d\phi}{|z\hat{a}_z - a\hat{a}_r|}$$



$\rho(\vec{x}') = \lambda$

$$\Phi = \frac{\lambda a}{4\pi\varepsilon_0 \sqrt{z^2 + a^2}} \underbrace{\int_0^{2\pi} d\phi}_{2\pi} = \frac{\lambda a}{2\varepsilon_0 \sqrt{z^2 + a^2}}$$

$$W_E(0) = \frac{1}{2} \cdot \frac{q\lambda a}{2\varepsilon_0 \sqrt{0^2 - a^2}} = \frac{q\lambda a}{4\varepsilon_0 |a|} = \boxed{\frac{q\lambda}{4\varepsilon_0}} \qquad a > 0$$

9-

$\vec{P} = P_0 \hat{a}_z$

$\vec{x} = z\hat{a}_z$

$\vec{x}' = r\hat{a}_r$

$dV = z\, r\, dr\, d\phi$

$$\Phi = \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho(\vec{x}')(\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} dV = \frac{hP_0}{4\pi\varepsilon_0} \int_V \frac{\hat{a}_z(z\hat{a}_z - r\hat{a}_r)}{|z\hat{a}_z - r\hat{a}_r|^3} dV$$

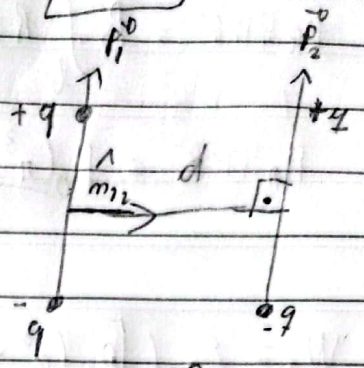
$$\Phi = \frac{zhP_0}{4\pi\varepsilon_0} \int_0^{2\pi} d\phi \int_0^a \frac{r\, dr}{(z^2 + r^2)^{3/2}} \Rightarrow \frac{hP_0 z}{2\varepsilon_0} \int_0^a \frac{r\, dr}{(z^2 + r^2)^{3/2}} \Bigg| \begin{array}{l} u = z^2 + r^2 \\ du = 2r\, dr \\ r\, dr = \frac{du}{2} \end{array}$$

$$\frac{1}{2} \int_{u_1}^{u_2} u^{-3/2}\, du = \frac{1}{\cancel{2}} \cdot (-\cancel{2}\sqrt{u}) = \frac{-1}{\sqrt{u}} = \frac{1}{\sqrt{z^2 + r^2}} \Bigg|_0^a$$

$$\Phi = \frac{hP_0 z}{2\varepsilon_0} \cdot \left[\frac{-1}{\sqrt{z^2 + a^2}} + \frac{1}{\sqrt{z^2}}\right] \Rightarrow \boxed{\frac{P_0 h a^2}{4\varepsilon_0 |z|^3}}$$

10 - $W = \frac{1}{2}\Phi q \Rightarrow \frac{1}{2}\frac{P_0 h a^2}{4\varepsilon_0 |z|^2}$, $W(h) = \frac{P_0 h a^2}{8\varepsilon_0 h^2} = \boxed{\frac{P_0 a^2}{8\varepsilon_0 h}}$

11 - $\vec{E} = \frac{1}{4\pi\varepsilon_0 d^3}\left[3(\vec{P}_1 \cdot \hat{n}_{12})\,\hat{n}_{12} - P_1\right]$



$\vec{E}_2 = \frac{1}{4\pi\varepsilon_0 d^3}\left[3(\cos(\frac{\pi}{2}) \cdot \hat{n}_{12} - P_1\right]$ 0

$\boxed{E_2 = -\frac{\vec{P}_1}{4\pi\varepsilon_0 d^3}}$

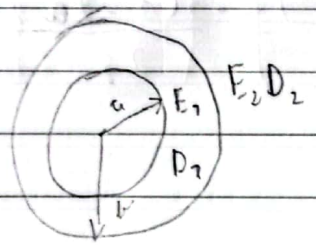
$U = -\vec{P}\vec{E}$

$\vec{E} = -\vec{\nabla}\Phi$

$90° = \cos(\frac{\pi}{2})$

$P_1 \cdot \hat{n}_{12} = 0$

12 -

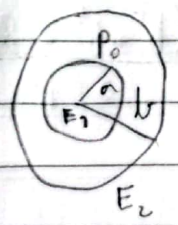


$\vec{E}_1 = \frac{P_n}{\varepsilon_0}$, $\vec{D}_1 = \varepsilon \vec{E}_1$

$E_2 = \frac{P_n}{\varepsilon_0}$, $\vec{D}_2 = \varepsilon_0 \vec{E}_2$

$\vec{E}_2$ não sofre efeito do dielétrico, já $\vec{E}_1$ sofre essa influência.

13 -



$W_E = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}\,dv \Rightarrow \frac{\varepsilon_0}{2}\int_V |E|^2 dv$

$\vec{E}_1 = \frac{P_0(a^3 - R^3)}{\varepsilon_0 3R^2}$, $a < R < b$

$\vec{E}_2 = \frac{P_0(R^3 - b^3)}{3R^2\varepsilon_0}$

$\oint \vec{D}_R \cdot d\vec{\varepsilon}_1 = \int P_0 dv$

$D_1 4\pi R^2 = P_0 \int_R^a R^2 dR \int_0^{4\pi} d\Omega$

$D_1 4\pi R^2 = \frac{P_0(a^3 - R^3)4\pi}{3} \Rightarrow D_1 = \frac{P_0(a^3 - R^3)}{3R^2}$, $a < R < b$

$\frac{4R^4}{4}$ $\frac{R^7}{7}$

$$\oint_{\Sigma_2} D_R \cdot d\Sigma_2 = \oint_V \rho_0 \, dv \qquad E_2 = \frac{\rho_0 (R^3 - b^3)}{3R^2 \varepsilon_0}$$

$$D_R 4\pi R^2 = \rho_0 \int_b^R R^2 dR \int_0^{4\pi} d\Omega$$

$$D_R 4\pi R^2 = \rho_0 \frac{(R^3 - b^3)}{3} 4\pi \Rightarrow D_R = \frac{\rho_0 (R^3 - b^3)}{3R^2}, \quad R \geq b$$

$$W_1 = \frac{\varepsilon_0}{2} \int_V \left[\frac{\rho_0 (a^3 - R^3)}{3R^2 \varepsilon_0}\right]^2 dv \Rightarrow \frac{\rho^2}{18} \int_V \frac{(a^3 - R^3)^2}{R^4} R^2 \sin\theta \, dR \, d\theta \, d\phi$$

$$\frac{\rho^2}{18} \int_0^a \frac{a^6 - 2a^3R^3 - R^6}{R^2} \int_0^{4\pi} d\Omega \Rightarrow \frac{2\pi\rho^2}{9} \int_0^a \frac{a^6 - 2a^3R^3 + R^6}{R^2} dR$$

$$W_1 = \frac{2\pi\rho^2}{9} \left[\frac{3(a^6 - 8a^3R^4 - R^7)}{28R^3}\right]_0^a \Rightarrow \left(\frac{3(a^6 - 8a^7 + a^7)}{28a^3}\right) = -\frac{3(a^6 - 7a^7)}{28a^3}$$

$$W_1 = \frac{6\pi\rho^2 (a^3 - 7a^4)}{252} = \boxed{\frac{\pi\rho^2 (a^3 - 7a^4)}{42}}$$

$$W_2 = \frac{\varepsilon_0}{2} \int_V \left[\frac{\rho_0 (R^3 - b^3)}{3R^2 \varepsilon_0}\right]^2 dv \Rightarrow \frac{\rho^2}{18} \int_0^b \frac{(R^3 - b^3)^2 R^2}{R^4} \int_0^{4\pi} d\Omega$$

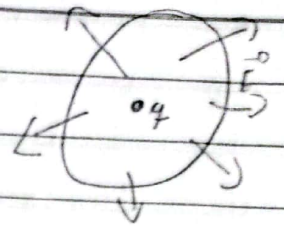
$$\frac{2\pi\rho^2}{9} \int_0^b \frac{R^6 + b^6 - 2b^3R^3}{R^2} dR \Rightarrow \left[\frac{3(R^7 + b^6 - 8b^3R^4)}{28R^3}\right]_0^b \Rightarrow \frac{3(b^7 + b^6 - 8b^7)}{28b^3}$$

$$W_2 = \frac{6\pi\rho^2}{252} \cdot (b^3 - 7b^4) \Rightarrow \boxed{\frac{\pi\rho^2 (b^3 - 7b^4)}{42}}$$

tilibra

CamScanner

14 → $\vec{E} = \frac{q}{4\pi\varepsilon_0 R^2}\hat{a}_R$, $\oint \vec{E}\cdot d\vec{\varepsilon} = \frac{q}{\varepsilon_0}$



$$\oint_{\varepsilon} \vec{E}\cdot d\vec{\varepsilon} = \frac{q}{\varepsilon_0}$$

$$\iint \vec{E}\cdot\hat{a}_R (R^2 \operatorname{sen}\theta \, d\theta \, d\phi)\,\hat{a}_R$$

$$\int_0^R \int_0^{4\pi} E R^2 \, d\Omega = \frac{q}{\varepsilon_0} \Rightarrow 4\pi \vec{E} \int_0^R R^2 \, dR = \frac{q}{\varepsilon_0}$$

$$4\pi \vec{E} R^2 = \frac{q}{\varepsilon_0} \Rightarrow \boxed{\vec{E} = \frac{q}{4\pi\varepsilon_0 R^2}}$$

15- $\oint_{\varepsilon} \vec{D}\cdot d\vec{\varepsilon} = \int_V \rho \, dv$, Lei de gauss para o Eletromagnetismo

gauss também afirma que:

$$\oint \vec{D}\cdot d\vec{\varepsilon} = \int_V \vec{\nabla}\cdot\vec{D}\, dv = \int \rho \, dv$$

$$= \int_V \vec{\nabla}\cdot\vec{D} - \rho \, dv = 0$$, dividindo dos dois lados temos

$$\vec{\nabla}\cdot\vec{D} - \rho = 0$$

$$\vec{\nabla}\cdot\vec{D} = \rho$$

$\oint_{\varepsilon} \vec{E}\cdot d\vec{\ell} = 0$, Lei de STOKES para o Eletromagnetismo

$$\oint_{\varepsilon} \vec{E}\cdot d\vec{\ell} = \int_{\varepsilon} \vec{\nabla}\times\vec{E}\, d\varepsilon = 0$$, $\vec{\nabla}\times\vec{E} = 0$ é uma propriedade do campo elétrico, ou seja ele é irrotacional

$$\vec{\nabla}\times\vec{E} = 0$$